ACADEMIC USE DISCLAIMER:

1. INTERPRET DRAWING IAW STANDARDS PRESCRIBED BY ASME Y14.100.
2. UNLESS OTHERWISE SPECIFIED:

   ALL RESISTORS ARE 1%, 0.1W, 0603

   ALL CAPACITORS ARE 10%, 16V, 0603
3. PARTIAL REFERENCE DESIGNATIONS ARE SHOWN FOR COMPLETE.
4. SEE TABLE A FOR POWER AND GROUND INPUTS TO MICROCIRCUITS.
5. INTERPRET REFERENCE DESIGNATIONS AS FOLLOWS:

   EXAMPLE: U1 - 2

   COMPONENT PIN NUMBER
6. GENERIC PART NUMBERS ARE SHOWN FOR REFERENCE ONLY.
7. CONNECT CHASSIS TO THE MOUNTING HOLES / CHASSIS GROUND.
8. LATEST REVISION CHANGES: NaN
9. UNUSED REFERENCES: R40, R41, R42, C28, C29, C30, C38, C39, C40

   → DO NOT INSTALL THESE COMPONENTS. LEAVE THEM UNPOPULATED.

REVISIONS

ZONE	LTR	DESCRIPTION	DATE	APPR.
	-	RELEASED PER ENGINEERING CSULB-DEESC	260207	T STOLPE

TABLE A

REFERENCE DESIGNATION	POWER INPUT TO PINS							
	PWR	SRC						
	+24 V	6S BATT						
	+10 V	PSU						
	+3.3 V	BUCK						
	GND	RTN						

INSTALLATION GUIDE:

1. MCU BOARD GOES ON TOP OF THE STACK SO USE FEMALE HEADERS WITH EXTENDED PINS. INSTALL 1X12 FEMALE HEADERS WITH EXTENDED PINS INTO P5 AND P6. P/N: 1568-14322-ND
2. IF USING HALL EFFECT SENSORS, INSTALL DRV5033 (TO-92 PACKAGE) BY SOLDERING WIRES DIRECTLY INTO THE SOLDER PADS AT P4. NOTE THE PHASES ARE LABELED AT P4 ON PCB. DO NOT INSTALL HEADER PINS HERE. IT MAY CAUSE COLLISIONS WITH THE STACK.
3. INSTALL P/N: 1568-14322-ND INTO P5 AND P6.
4. P/N PPPC032LJBN-RC MAY BE INSTALLED INTO P1 FOR EASE TO PROGRAM. TAKE NOTE OF THE PIN DEFINTIONS AT P1 ON SILK SCREEN BEFORE INSTALLATION. REFER TO DOCUMENTATION IF THE SILK SCREEN LABELING IS COVERED.
5. THIS BOARD ALLOWS FOR AN EXTERNAL 3.3 V SOURCE TO BE USED IN PLACE OF BUILT-IN BUCK CONVERTERS IN THE DRV8323RSRGZR CHIPS. USE P7 TO SOLDER WIRES DIRECTLY INTO THE THROUGH HOLE PADS. FOLLOW LABELING ON PCB.
6. USE P2 TO INSTALL A POTENTIOMETER AND SW. USE P2_1, P2_3, P2_5 PINS FOR POT. POTENTIOMETER CAN BE USED TO CONTROL SPEED CONTROL. NO POTENTIOMETER CAN BE INSTALLED AND SPEED CONTROL MAY BE STRICTLY SOFTWARE DEFINED. INSTALL A SPST SWITCH INTO P4_4 AND P4_6. THIS SWITCH MAY BE USED TO CONTROL THE DIRECTION OF THE BLDC MOTOR.
7. INSTALL 1X4 HEADER PINS INTO P8 WITH EXTENDED PINS. P/N: SAM1198-04-ND THESE HEADERS WILL EXTEND THROUGH EVERY BOARD IN THE STACK TO CARRY A CAN BUS UP THE STACK.
8. P3 ALLOWS THE OPTION TO CONNECT A CC1101 RF MODULE CONFIGURED TO THE 433MHZ FREQUENCY FOR COMMUNICATIONS TO ANY DEVICE OPERATING ON 433MHZ. IF MULTIPLE BOARDS ARE GETTING STACKED, PLEASE USE THE TOP-MOST BOARD FOR THE RF MODULE. DO NOT INSTALL HEADER PINS ON ANY BOARDS UNDER THE TOP-MOST BOARD IN THE STACK. REFER TO ARROWS ON SILKSCREEN FOR INSTALLATION ORIENTATION
9. MOUNTING HOLES ARE SIZED FOR M3 SCREWS. MOUNTING HOLES ARE PLATED WITH COPPER AND ARE CONNECTED TO GROUND ON THE MCU BOARD.

NOTICE OF OWNERSHIP AND INTELLECTUAL PROPERTY

CONTRACT NO	EE400D
DWN BY	T STOLPE
CHECK BY	J TOLDEN
APPV BY	J TOLDEN
PROF	DR. JUFOUT
MKS Rev / MKS Date	THOMAS STOLPE, JON TOLDEN, BAO PHAM, AUTUMN AGUILAR

CALIFORNIA STATE UNIVERSITY - LONG BEACH
ELECTRICAL ENGINEERING
LONG BEACH, CALIFORNIA

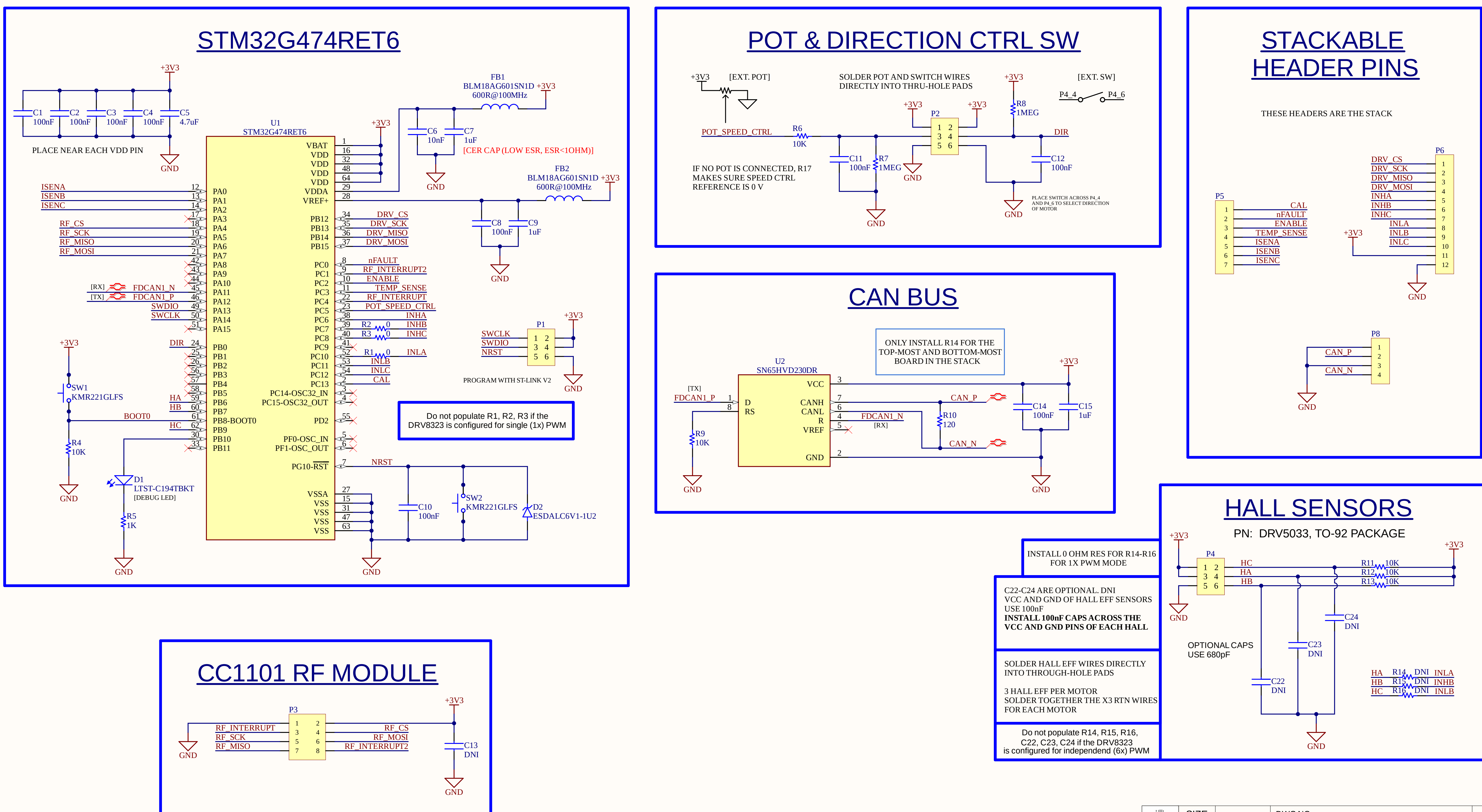
TITLE

MCU CONTROL FOR ESC

SIZE		DWG NO	REV
B	SPRING 2026 CSULB EE400D SENIOR DESIGN	F-65-50505	-
SCALE: NONE	VIEWDRAW	SHEET 1 OF 2	

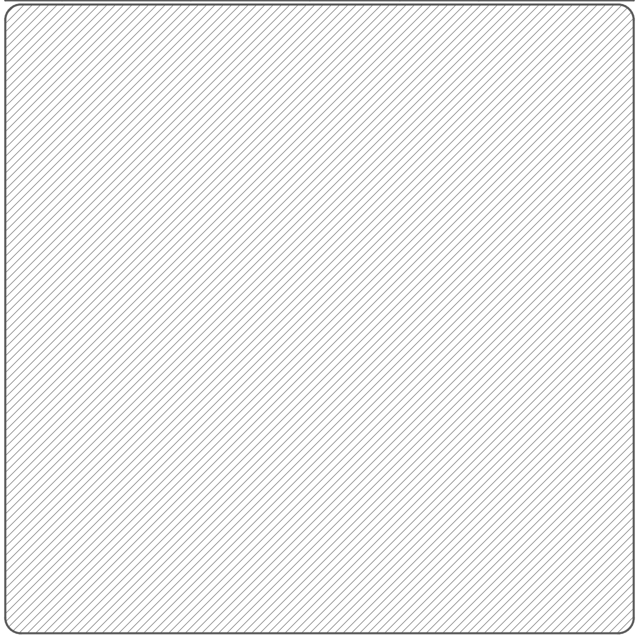
MCU AND PERIPHERALS

UNLESS OTHERWISE NOTED:
ALL RESISTORS ARE 1%, 0.1W
ALL CAPACITORS ARE 10%, 50V

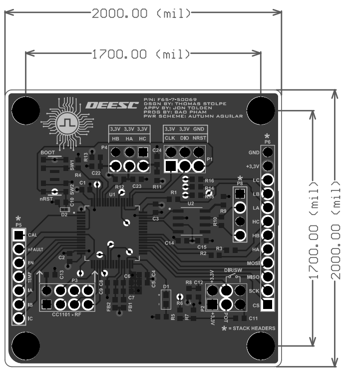


SIZE	SPRING 2026 CSULB EE400D SENIOR DESIGN	DWG NO	REV
B		MCU_1	-
SCALE: NONE	VIEWDRAW		SHEET 2 OF 3

Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.010mm	3.5	
1	LAYER1		0.036mm		
	Dielectric 1	PP-006	0.178mm	4.1	
2	LAYER2	CF-004	0.035mm		
	Dielectric 2	FR-4	1.001mm	4.8	
3	LAYER3	CF-004	0.035mm		
	Dielectric 3	PP-006	0.178mm	4.1	
4	LAYER4		0.036mm		
	Bottom Solder	Solder Resist	0.010mm	3.5	
	Bottom Overlay				



LAYER 1: SIGNAL
LAYER 2: GND
LAYER 3: +3.3 V
LAYER 4: SIGNAL



DESIGN INFORMATION

MIN. TRACK WIDTH: 10 MIL
MIN. CLEARANCE: 6 MIL
MIN. VIA PAD SIZE: 20 MIL
MINIMUM ANNULAR RING 0.05mm (2MIL) EXTERNAL
PER IPC-D-275 CLASS 2 LEVEL C
REGISTRATION TOLERANCES: METAL +/- 5 MIL, HOLES +/- 3 MIL
HOLE SIZE TOLERANCE (UNLESS OTHERWISE SPECIFIED): +/- 3 MIL

MATERIAL:
[ ] FR-408 [X] FR-4 High Tg [ ] OTHER ______
THICKNESS: [X] 62 MIL (1.6mm) +/-10% [ ] OTHER ______
TOLERANCE: [X] ANSI IPC-6012 TYPE 3 CLASS 2
[ ] OTHER +/- ______
BOW & TWIST: [X] ANSI IPC-6012 TYPE 3 CLASS 2
[ ] OTHER +/- ______

DRILLING:
REFERENCE: [X] AS SHOWN [X] NC_DRILL FILES
PTH COPPER THICKNESS: [X] 20-30 um [ ] OTHER ______

BOARD FINISH:
SILKSCREEN: [X] TOP [X] BOTTOM
SILKSCREEN COLOR: [X] WHITE [ ] OTHER ______
SOLDER RESIST COLOR: [ ] GREEN [X] OTHER RED
[X] MATTE [ ] SEMI-GLOSS

SURFACE FINISH: [X] IMMERSION GOLD (ENIG) [ ] ENEPIG
[ ] IMM. TIN/SILVER OR EQUIV [ ] OTHER ______

ARRAY/PANEL: [ ] CUT AND TRIM PER M1 BOARD OUTLINE
[ ] N.C. ROUTE [X] V. SCORE

CERTIFICATION: MATERIALS AND WORKMANSHIP FOR ALL PCBS TO MEET OR EXCEED THE REQUIREMENTS OF:
[X] ANSI IPC-A-600F CLASS -> [ ] 1 [X] 2 [ ] 3
[X] RoHS [ ] OTHER PER ORDER

ALL BOARDS MUST MEET OR EXCEED EE400D REQUIREMENTS.

ADDITIONAL REQUIREMENTS:
MICROSECTION: [ ] YES
BARE BOARD ELEC. TEST: [ ] NONE [X] REQUIRED [ ] PER ORDER

DEESC

TEAM LEAD: JON TOLDEN
THOMAS STOLPE
BAO PHAM
AUTUMN AGUILAR

PROJECT TITLE:
MCU BOARD FOR ESC

DESIGNED FOR:
CSULB EE400D SENIOR PROJECT

FILE NAME:
MCU_EE400D_PCB.PcbDoc

1000.00mil

COMPONENTS MARKED 'DNI' SHOULD NOT BE POPULATED.
ASSEMBLY VARIANT: [No Variations]

PCB VIEWED FROM TOP SIDE

GENERATED : 8/7/2026 8:52:44 PM

PROJECT MANAGER:
JON TOLDEN

LAYOUT BY:
THOMAS STOLPE

SCALE: 0.72

ALTIUM DESIGNER VERSION:
26.8.1.31